

FlowForge — Full-Stack Engineering Exercise

Time-box: 3–4 hours • Stack: TypeScript, Express, React, React Flow

Scenario

You have joined a team building **FlowForge**, an early prototype of a node-based automation builder. Users drag *Input*, *Process*, and *Output* nodes onto a canvas, connect them, and run the flow so data passes from one node to the next. Flows are saved to disk as JSON. The prototype works, but it was written quickly and now needs an engineer's care.

What the app does today

- Add nodes from the toolbar: **Input** (a manual value), **Transform** (rewrites its input via a `{{input}}` template, with an uppercase option), and **Log** (shows the result).
- Connect nodes, click **Run** to execute, and **Save / Reload** to persist and restore a flow.

How to run

See `README.md`. You need Node.js 18+. Start the backend and the frontend, then open the app in your browser.

Your tasks

1. Improve the codebase

The prototype runs but is not structured for a team to build on. Leave it in a state you would be comfortable handing to a colleague. Behavior must not regress.

2. Add an HTTP / Webhook output node

Add a new output node that sends the processed result to an external endpoint when the flow runs. It must support a configurable destination URL and request method, and the outcome of the request must be visible in the app. Choose sensible defaults.

3. Find and fix the bugs

While using the app you will notice behavior that is incorrect. There are one or two such defects. Reproduce them, fix them, and make sure they stay fixed.

What we expect

- **Correctness:** the app runs; your feature works end to end; the bugs are gone.
- **Code quality:** clear structure, sensible boundaries, no needless duplication, reasonable types.
- **Git hygiene:** incremental, well-described commits.
- **Communication:** a short `NOTES.md` describing your key decisions, trade-offs, and anything you would do with more time.

Out of scope

Authentication, deployment, a real database, and visual/design polish. A JSON file for storage is fine. Do not spend time on these.

Submission

Submit as a git repository (or a zip that includes the `.git` history). Keep the app runnable via the steps in `README.md`.